

Surabhi Chavan

+1 (619) 666 2309 | San Diego, CA | schavan2264@sdsu.edu | [LinkedIn](#) | [GitHub](#) | [Medium](#) | [Portfolio](#)

EDUCATION

Master of Science in Information Systems

San Diego State University, USA | Aug 2024 - May 2026 | GPA: 3.5

Bachelor of Engineering

Savitribai Phule Pune University, India | May 2018 - Apr 2022 | GPA: 3.8

EXPERIENCE

Software Development Intern

Jan 2026 - Present

Donald Hans

Los Angeles, CA

- Built LLM-based data transformation pipeline on Cloudflare Workers, enabling low-latency edge inference via OpenRouter API.
- Implemented vector embedding pipeline for Retrieval-Augmented Generation (RAG) by generating embeddings, storing them in a vector index, and enabling semantic search for chatbot responses.
- Improved pipeline reliability by 30% by implementing async orchestration, retry logic, and multi-environment debugging.

Tech stack: React, Cloudflare, REST APIs, Agentic AI, SEO Optimization

Software Research Assistant

Jan 2025 - Dec 2025

AI4Business - San Diego State University

San Diego, CA

- Reduced LLM hallucination by 14% by developing a novel approach leveraging Agentic AI systems and AutoGen frameworks.
- Improved handwriting recognition accuracy to 92% by designing a supervised ML model trained on diverse datasets.
- Achieved 90% image classification accuracy by fine-tuning Deep Learning models (AlexNet CNN) in MATLAB.
- Enhanced email security by 30% through an NLP phishing-detection model that analyzes email pattern anomalies.

Tech stack: Python, PySpark, MATLAB, TensorFlow, Scikit-learn, AutoGen, Agentic AI, NLP, Data Analysis, Model Evaluation

Software Engineer

Sep 2022 - Aug 2024

JSat Automation Inc.

India

- Built a scalable scheduling system with Angular, REST APIs, and AWS Lambda/API Gateway for high-availability planning.
- Optimized real-time system performance by reducing latency 35% and improving usability 60% via TypeScript/RxJS reactive pipelines, refining async execution paths, and using CloudWatch metrics.
- Reduced build times by 40% through Gulp automation for transpilation, CSS preprocessing, and image pipeline enhancements.
- Boosted PostgreSQL query performance by 45% through indexing, plan tuning, and refined connection-pool management.
- Improved scalability by deploying Dockerized microservices with centralized logging and environment-specific configurations.
- Designed e-workflow and scheduling UX prototypes in Figma, enabling cross-functional stakeholder alignment before development, reducing rework cycles by 25%.

Tech stack: JavaScript (ES6+), TypeScript, Angular, RxJS, Node.js, REST APIs, PostgreSQL, Three.js, WebGL, Docker, AWS Lambda/API Gateway, S3, CloudWatch, Gulp, Figma, Git, Agile Development, CI/CD, UI/UX Design

Software Engineer Intern

Mar 2022 - Aug 2022

Qualys Security

India

- Improved vulnerability detection accuracy by 40% by automating CVE-CPE mapping through CISA API integration and repository synchronization.
- Built automation using the Shodan API to identify affected hosts and IPs, strengthening external attack surface monitoring.
- Developed scripts to compare technology repositories and generate CSV reports for missing CPEs and outdated entries.

Tech stack: Java, Spring Boot, REST APIs, JSON, Python, Oracle DB, Shodan API, CVE/CPE Mapping, Vulnerability Management

PROJECTS

ROOM | Video Conferencing Platform | React, Next.js, TypeScript, Clerk, getstream, shadcn, Tailwind CSS

- Built a real-time video conferencing platform using Next.js 14 and TypeScript for high performance and scalability.
- Integrated Stream Video SDK for live audio/video streaming and Clerk for secure authentication and user management.
- Delivered core features like instant meetings, scheduling, recordings, and join-via-ID to enhance usability.
- Designed an accessible, modern UI using shadcn/ui and Radix UI, enabling seamless collaboration for remote teams.

Bug Tracker | TypeScript, Node.js, Express, JWT Auth, RESTful API

- Built a TypeScript/Express bug-tracking API with JWT auth, role-based access control, and secure signed URL file downloads.
- Designed GET endpoints with nested collections, computed metrics, pagination, and selective expansion via query params.
- Implemented idempotent PUT updates with field-level validation, audit logging, and custom admin-only escalation rule.
- Developed secure multipart file upload with size/type limits, expiring download tokens, and request-level trace logging for observability.